PROJECT: Multi-AZ WordPress Deployment

Services: EC2, RDS (MySQL), EFS, ALB, Route 53, CloudFront

Goal: Highly available WordPress deployment with content delivery optimization.

Skills: Shared storage (EFS), DNS, caching strategies.

Definition: This project demonstrates the deployment of a highly available WordPress website on Amazon Web Services (AWS) using a multi-AZ architecture. The infrastructure integrates services such as Amazon EC2, Amazon RDS, Amazon EFS, Application Load Balancer (ALB), Route 53, and CloudFront to ensure scalability, reliability, and optimized content delivery. It showcases how cloud-native services can be combined to build a resilient and production-ready web application.

Scope of the Project: The project focuses on designing and implementing a fault-tolerant and scalable WordPress hosting environment in AWS. It covers configuring compute resources, managed databases, shared storage, DNS management, and global content delivery for improved performance. The implementation demonstrates real-world cloud architecture practices used to maintain high availability, load balancing, and efficient content distribution for web applications.

Methodologies:

1. **Multi-AZ WordPress Deployment on AWS:** Method used in this Project

- In this method, WordPress runs on Amazon EC2 instances with shared storage (EFS), while the database is managed using Amazon RDS (MySQL).
- Traffic is distributed using an Application Load Balancer, and Route 53 and CloudFront are used for DNS management and faster content delivery.
- This method is best because it provides high availability, scalability, and better performance. If one server or availability zone fails, the system can still continue running using other resources.
- Services like CloudFront and ALB also improve website speed and handle large amounts of traffic efficiently.

2. **Serverless Deployment:** Alternative Method

- In a serverless deployment, the application runs without managing servers by using services like AWS Lambda, API Gateway, and Amazon S3. The backend logic is executed through Lambda functions, while API Gateway handles incoming requests from users.
- Static website content such as HTML, CSS, and images can be hosted on Amazon S3. This method automatically scales based on traffic and reduces infrastructure management, but it may not be ideal for traditional applications like WordPress without additional configuration.

Limitations of Serverless Deployment

- ➢ Not ideal for traditional applications like WordPress because WordPress requires a persistent server environment and file system.
- ➢ Cold start delays can occur when Lambda functions are invoked after a period of inactivity.
- ➢ Execution time limits exist for Lambda functions, which may restrict long-running processes.
- ➢ More complex architecture and configuration compared to simple server-based deployments.

Services Used in this Project:

1. **Amazon EC2 (Elastic Compute Cloud)**

- **Definition:** Amazon EC2 is a cloud computing service that provides virtual servers to run applications on AWS.
- **Purpose:** EC2 is used to host and run the WordPress website with Apache and PHP.
- **Role in this Project:** EC2 acts as the web server that handles user requests and delivers the WordPress website.

2. **Amazon RDS (Relational Database Service)**

- **Definition:** Amazon RDS is a managed database service that allows users to easily create, manage, and maintain relational databases in the cloud.
- **Purpose:** Amazon RDS is used to store and manage the WordPress MySQL database.
- **Role in this Project**: Amazon RDS acts as the backend database that stores website data such as posts, users, and settings.

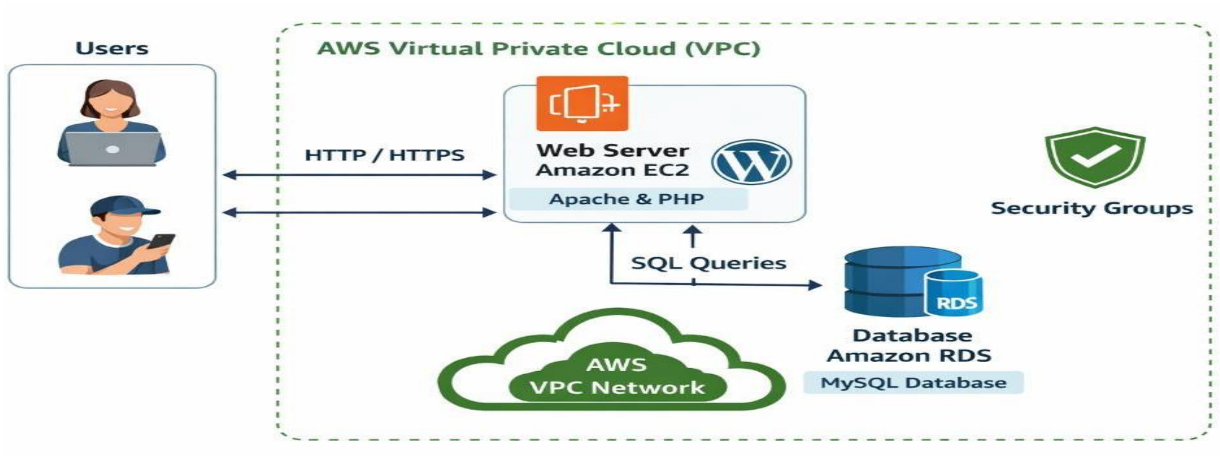
3. **Security Group**

- **Definition:** A Security Group is a virtual firewall in AWS that controls inbound and outbound network traffic for resources like EC2 and RDS.
- **Purpose:** The Security Group is used to allow necessary access such as HTTP, SSH, and database communication.
- **Role in this Project:** The Security Group ensures secure communication between the EC2 instance and the RDS database while protecting the server from unauthorized access.

4. **VPC (Virtual Private Cloud)**

- **Definition:** A VPC (Virtual Private Cloud) is a virtual network in AWS where cloud resources like EC2 and RDS are securely deployed.
- **Purpose:** The VPC provides the network environment for hosting the EC2 instance and RDS database.
- **Role in this Project:** The VPC enables secure communication between EC2 and RDS within the same network.

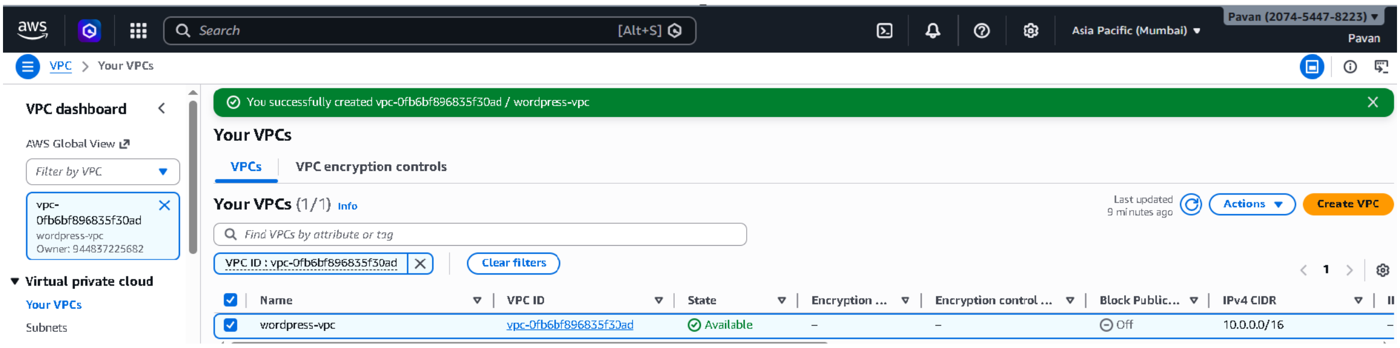
Architecture Diagram:



Implementation and Execution of Multi-AZ WordPress Deployment.

STEP 1: Create VPC

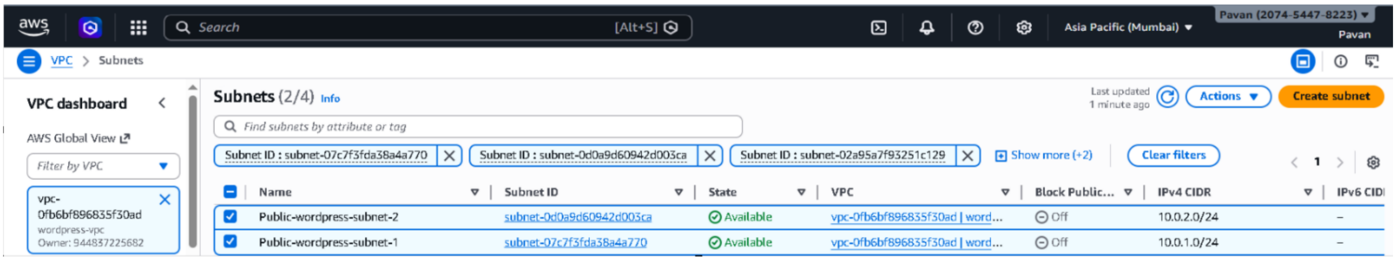
- Name: wordpress-vpc
- IPV4 CIDR: 10.0.0.0/16
- Click create vpc



STEP 2: Create Subnets

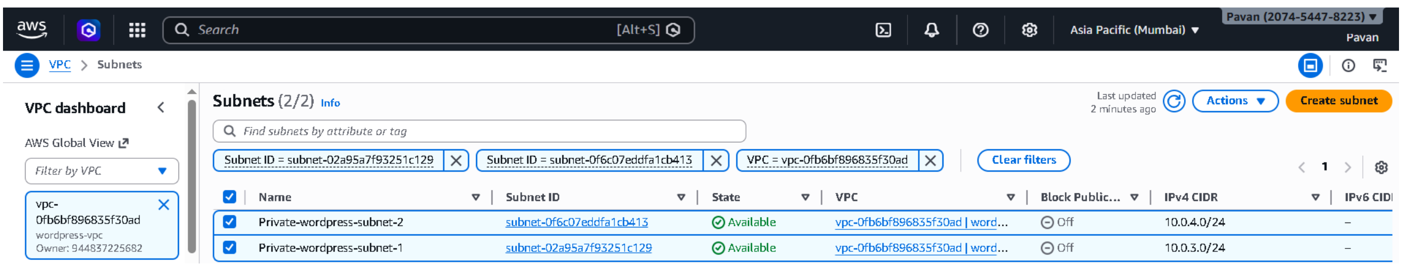
Public Subnets

- Select vpc: wordpress-vpc
- Subnet name: public-wordpress-subnet-1 and public-wordpress-subnet-2
- Availability Zone: ap-south-1a and ap-south-1b
- IPV4 CIDR: 10.0.1.0/24 and 10.0.2.0/24
- Click create subnet



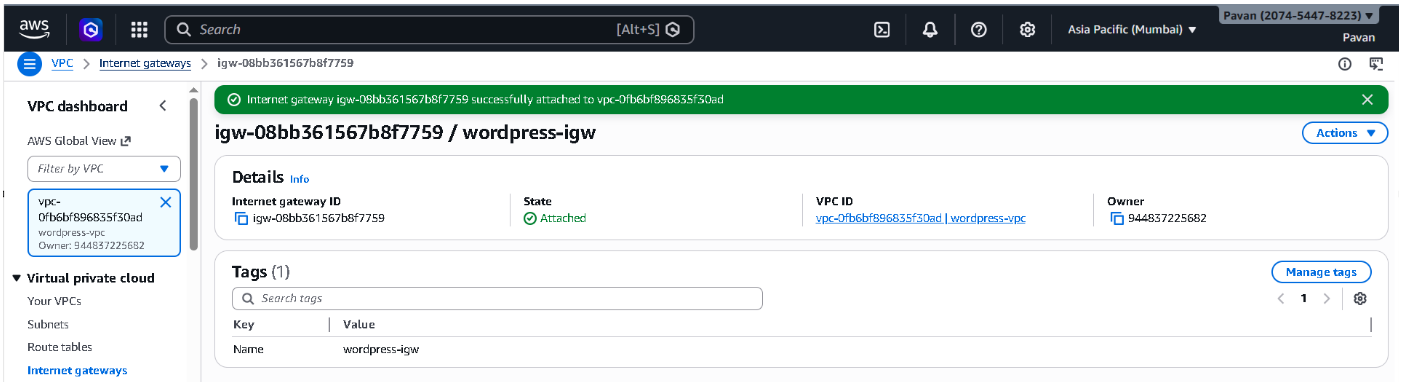
STEP 3: Private Subnets

- Select vpc: wordpress-vpc
- Subnet name: private-wordpress-subnet-1 and private-wordpress-subnet-2
- Availability Zone: ap-south-1a and ap-south-1b
- IPV4 CIDR: 10.0.3.0/24 and 10.0.4.0/24
- Click create subnet



STEP 4: Create Internet Gateway

- ➢ Name: wordpress-igw
- ➢ Click Create Internet Gateway
- ➢ Select IGW: Actions: Attach to VPC
- ➢ Click on Attach

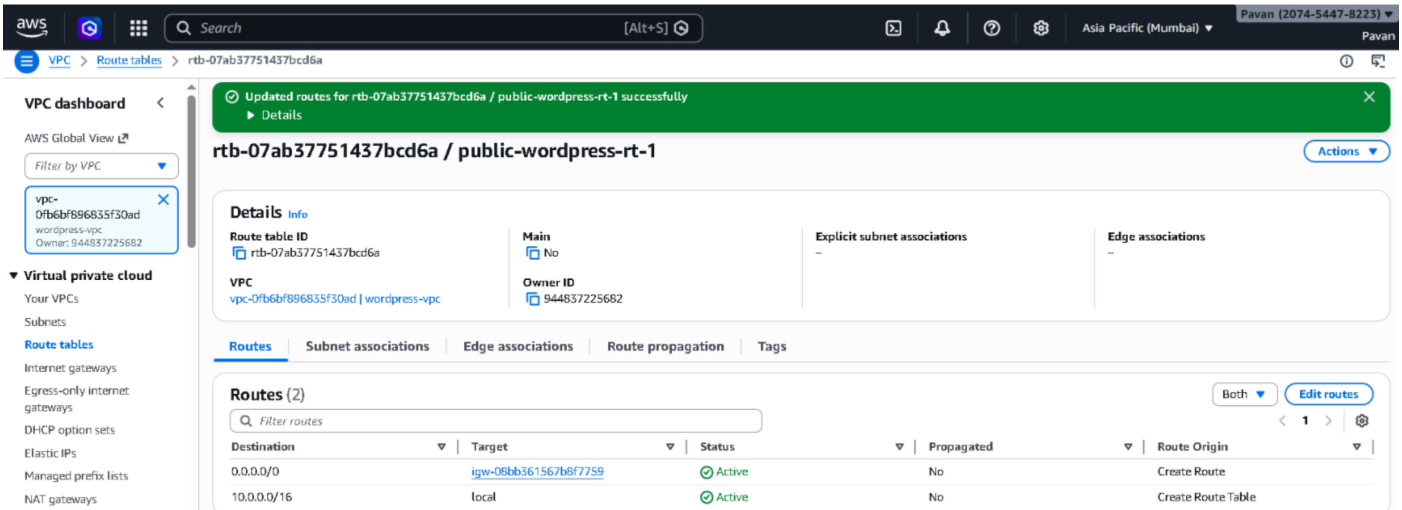


STEP 5: Create Public Route Table

- Name: public-wordpress-rt-1
- VPC: wordpress-vpc
- Click create route table

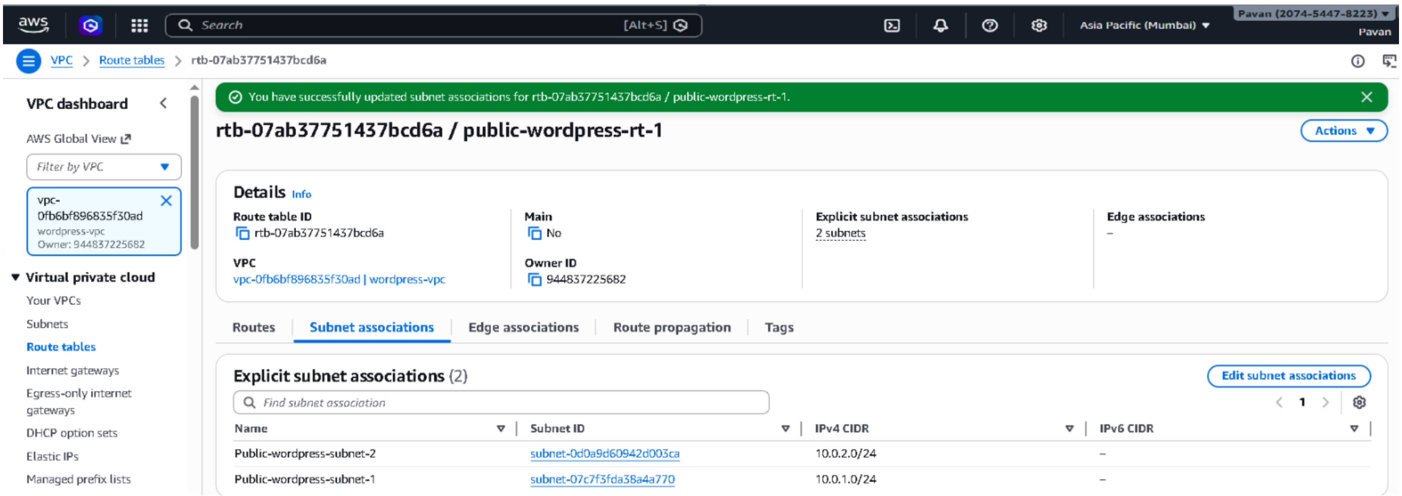
Add Route:

- Select public-wordpress-rt-1
- Routes: Edit routes
- Add routes:
- ➢ Destination: 0.0.0.0/0
- ➢ Target: Internet Gateway(wordpress-igw)
- ➢ Save routes

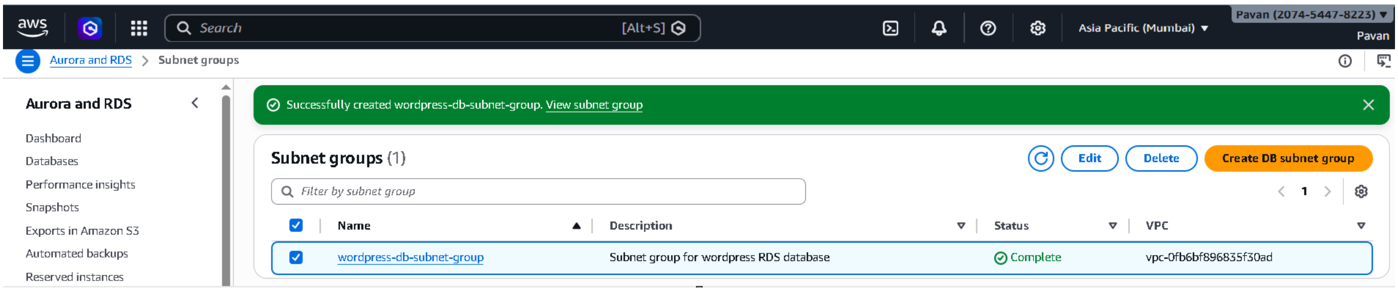


Associate Public Subnet:

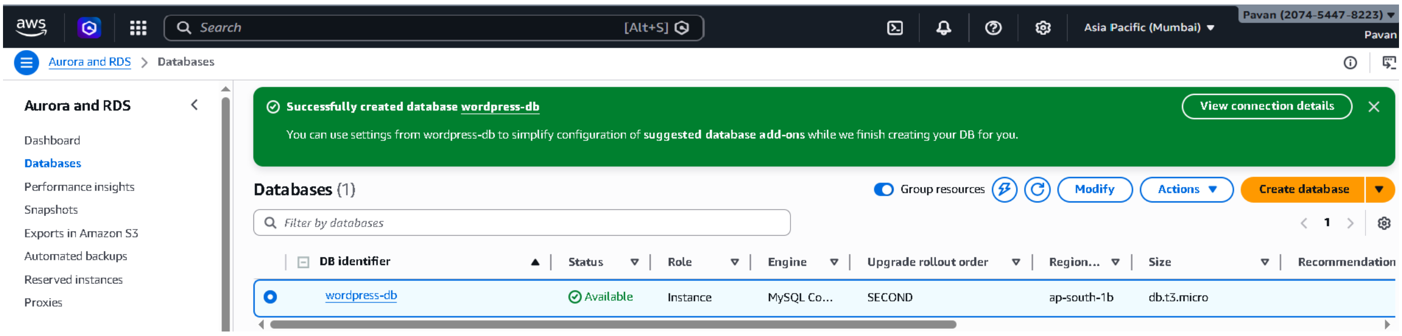
- Subnet Associations: Edit
- Select public-wordpress-subnet 1 and public-wordpress-subnet-2
- Save



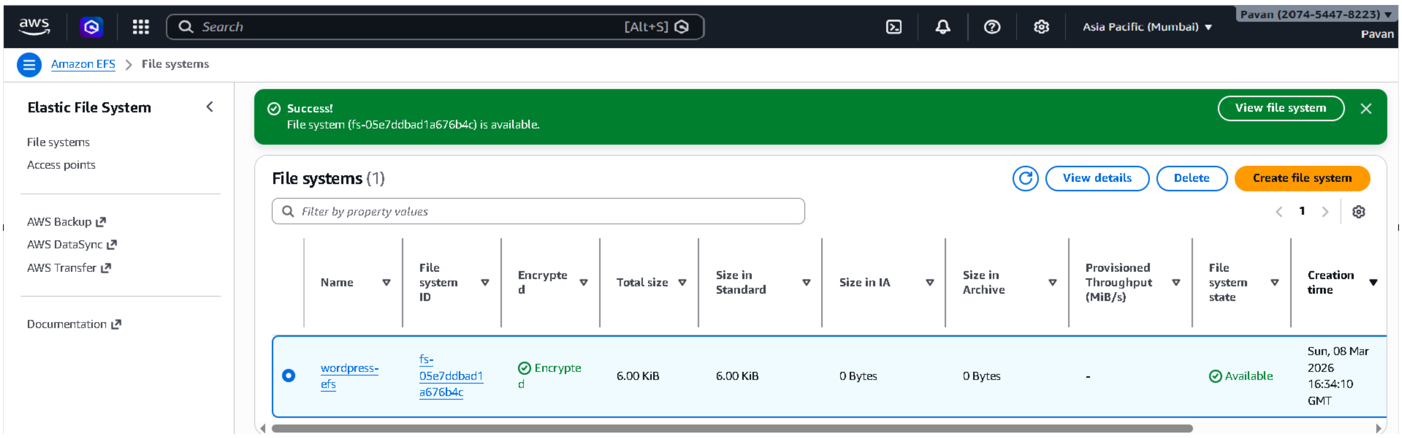
STEP 6: Created Subnet Group



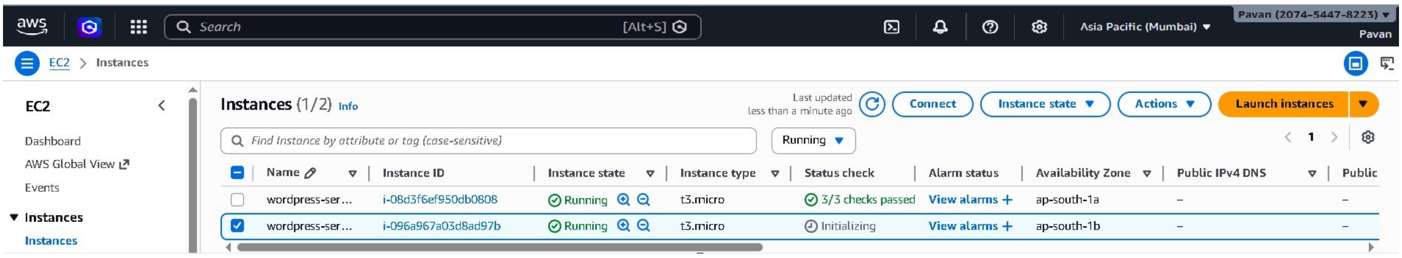
STEP 7: Created Database



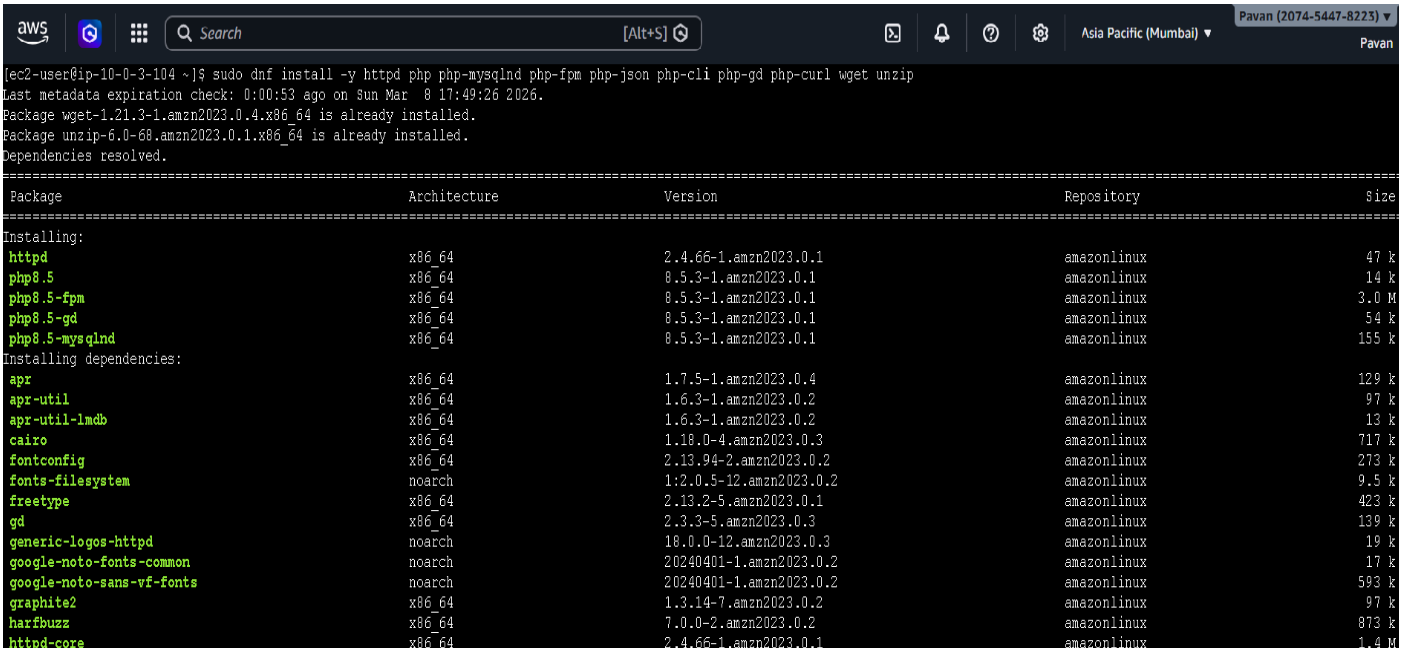
STEP 8: Created Elastic File System



STEP 9: Launch an instance



STEP 10: Update the server and Install Apache + PHP + MySQL client



STEP 11: Start Apache and enable

```
[ec2-user@ip-10-0-3-104 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-3-104 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-3-104 ~]$
```

STEP 12: Test Apache

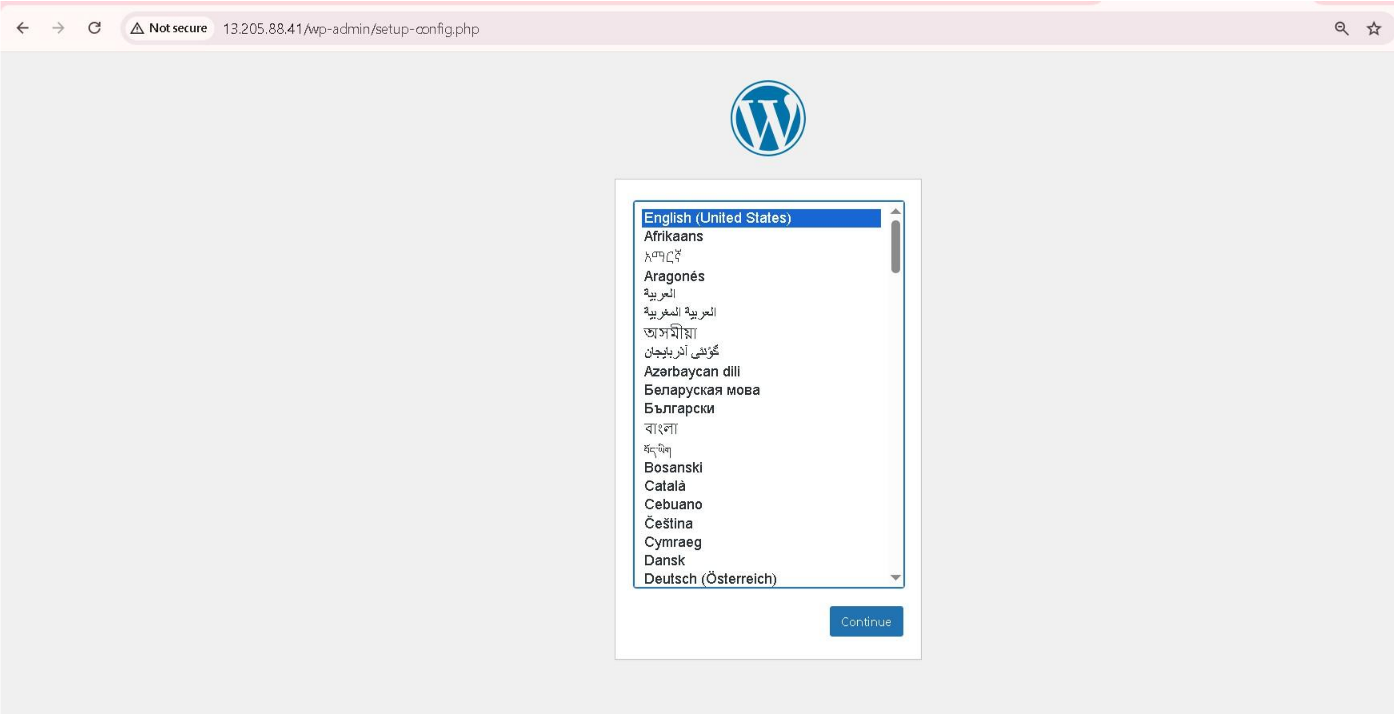
STEP 13: Download wordpress

```
[ec2-user@ip-10-0-3-104 html]$ sudo wget https://wordpress.org/latest.tar.gz
--2026-03-08 17:55:45--  https://wordpress.org/latest.tar.gz
Resolving wordpress.org (wordpress.org)... 198.143.164.252, 2607:f978:5:8002::c68f:a4fc
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27062929 (26M) [application/octet-stream]
Saving to: 'latest.tar.gz'

latest.tar.gz                    100%[==========================================================>]  25.81M  8.05MB/s    in 3.9s

2026-03-08 17:55:50 (6.63 MB/s) - 'latest.tar.gz' saved [27062929/27062929]

[ec2-user@ip-10-0-3-104 html]$
```

STEP 14: Restart Apache

```
[ec2-user@ip-10-0-3-104 html]$ sudo systemctl restart httpd
[ec2-user@ip-10-0-3-104 html]$
```

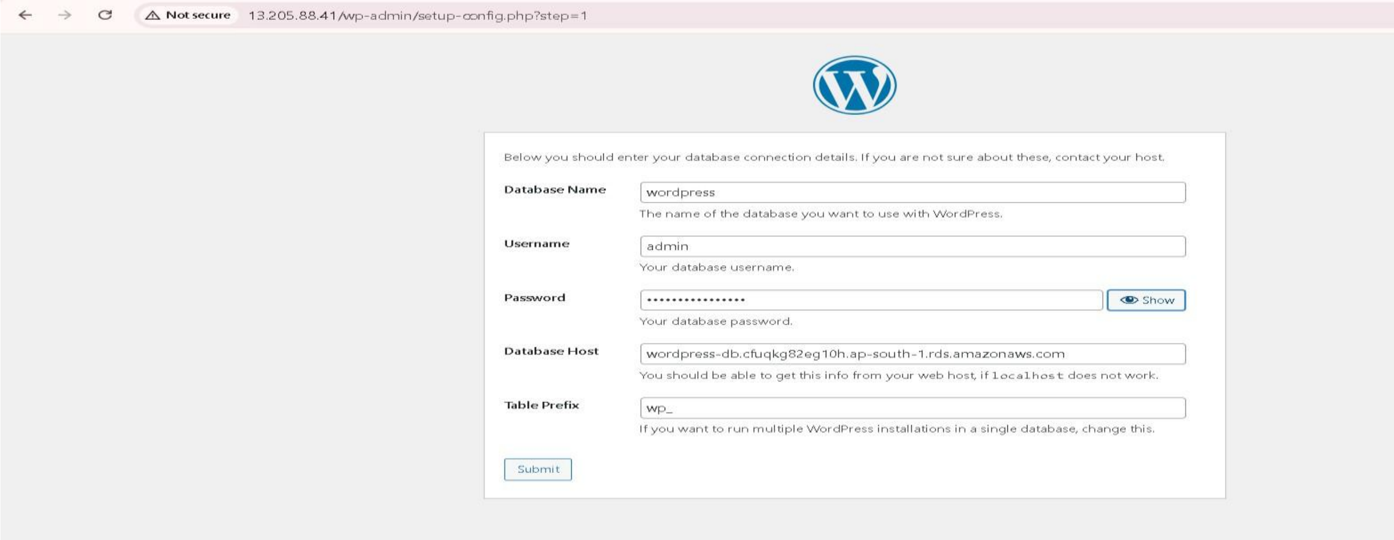
STEP 15: We can see wordpress installation page

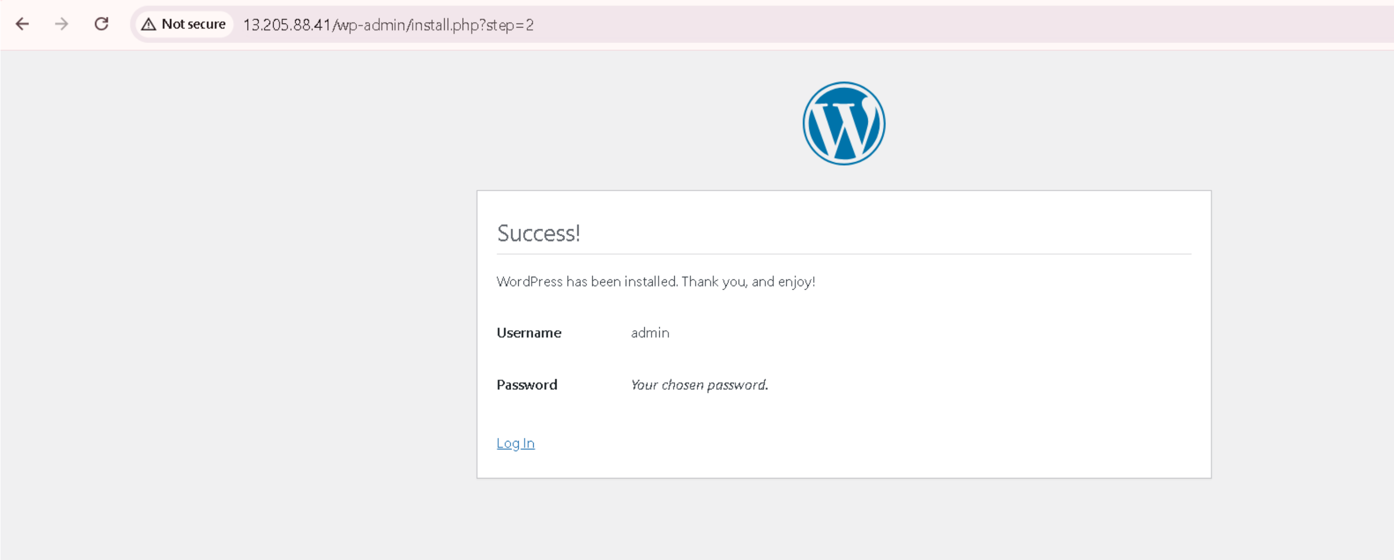


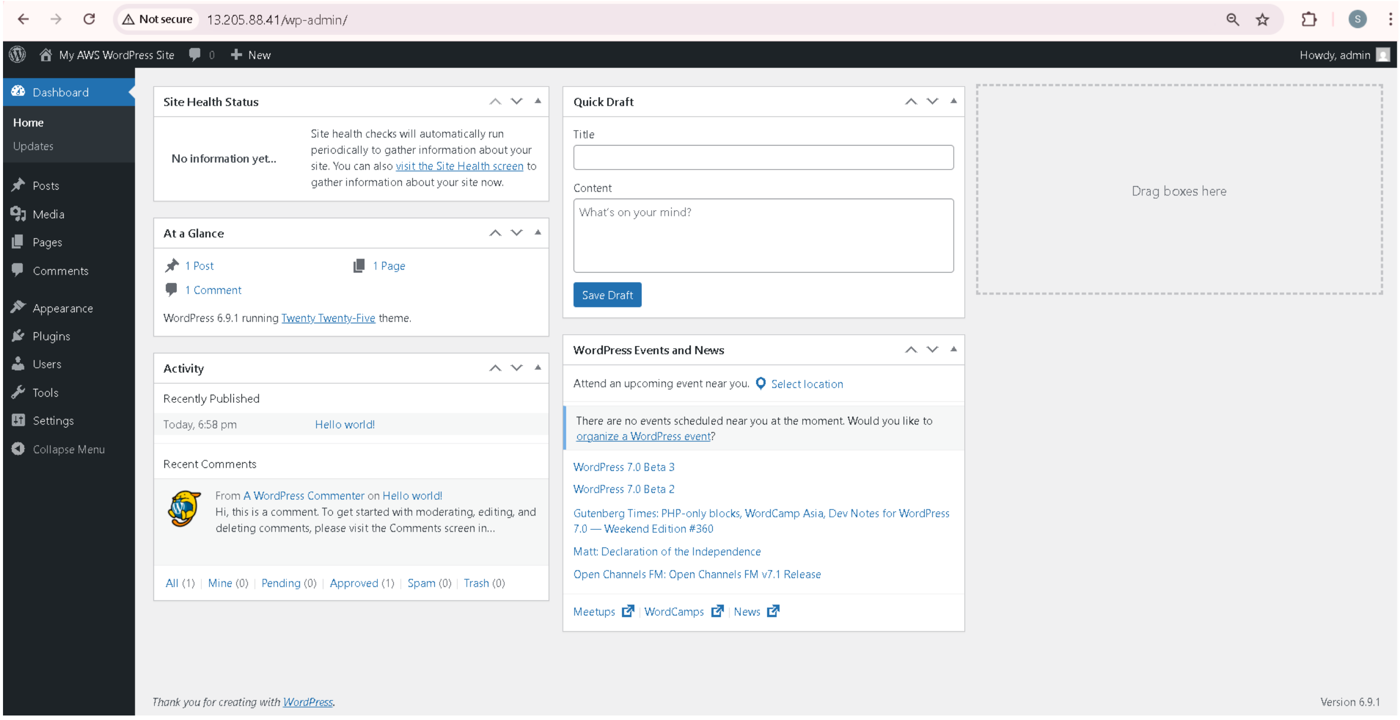
STEP 16: We will connect wordpress-→ RDS database

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Pavan (2074-5447-8223) | Pavan

```
[ec2-user@ip-10-0-3-104 ~]$ mysql -h wordpress-db.cfuqkg82g10h.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 106
Server version: 8.4.7 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> CREATE DATABASE wordpress;
Query OK, 1 row affected (0.018 sec)

MySQL [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| wordpress          |
+--------------------+
5 rows in set (0.013 sec)

MySQL [(none)]>
```


Not secure
13.205.88.41/wp-admin/setup-config.php?step=1
Below you should enter your database connection details. If you are not sure about these, contact your host.
Database Name
wordpress
The name of the database you want to use with WordPress.
Username
admin
Your database username.
Password
Show
Your database password.
Database Host
wordpress-db.cfuqkg82eg10h.ap-south-1.rds.amazonaws.com
You should be able to get this info from your web host, if localhost does not work.
Table Prefix
wp_
If you want to run multiple WordPress installations in a single database, change this.
Submit


Not secure
13.205.88.41/wp-admin/install.php?step=2
Success!
WordPress has been installed. Thank you, and enjoy!
Username
admin
Password
Your chosen password.
Log In


Not secure
13.205.88.41/wp-admin/
My AWS WordPress Site
0
New
Howdy, admin
Dashboard
Home
Updates
Posts
Media
Pages
Comments
Appearance
Plugins
Users
Tools
Settings
Collapse Menu
Site Health Status
No information yet...
Site health checks will automatically run periodically to gather information about your site. You can also visit the Site Health screen to gather information about your site now.
At a Glance
1 Post
1 Page
1 Comment
WordPress 6.9.1 running Twenty Twenty-Five theme.
Activity
Recently Published
Today, 6:58 pm
Hello world!
Recent Comments
From A WordPress Commenter on Hello world!
Hi, this is a comment. To get started with moderating, editing, and deleting comments, please visit the Comments screen in...
All (1) | Mine (0) | Pending (0) | Approved (1) | Spam (0) | Trash (0)
Quick Draft
Title
Content
What's on your mind?
Save Draft
WordPress Events and News
Attend an upcoming event near you.
Select location
There are no events scheduled near you at the moment. Would you like to organize a WordPress event?
WordPress 7.0 Beta 3
WordPress 7.0 Beta 2
Gutenberg Times: PHP-only blocks, WordCamp Asia, Dev Notes for WordPress 7.0 — Weekend Edition #360
Matt: Declaration of the Independence
Open Channels FM: Open Channels FM v7.1 Release
Meetups
WordCamps
News
Drag boxes here
Thank you for creating with WordPress.
Version 6.9.1

Troubleshooting:

1. WordPress Database Connection Error

- **Issue:** WordPress showed "Error establishing database connection" during setup.
- **Solution:** Updated the wp-config.php file with the correct database details:

- DB Name
- DB Username
- DB Password
- RDS Endpoint

2. Unable to Connect EC2 to RDS Database
    - **Issue:**
      The EC2 instance could not connect to the MySQL database hosted on Amazon RDS.
    - **Solution:**
      Added a MySQL rule (Port 3306) in the RDS security group and allowed access from the EC2 security group.

Conclusion:

This project successfully demonstrated the deployment of a WordPress website using Amazon EC2 and Amazon RDS on AWS cloud infrastructure. The implementation showed how cloud services can be used to build a scalable, reliable, and cost-effective web hosting environment. Through this project, we gained practical experience in cloud computing, server configuration, database connectivity, and security management. The architecture allows the website to be easily managed and accessed publicly through the internet. Overall, the project highlights how AWS services can simplify the process of deploying and managing modern web applications.